Microsoft Elevate

CAPSTONE PROJECT

HEART ATTACK RISK ANALYSIS DASHBOARD

PRESENTED BY

STUDENT NAME: KRUSHNA VITTHAL RATHOD

COLLEGE NAME: YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING

DEPARTMENT: AI & DATA SCIENCE

EMAIL ID: krushnarathod0077@gmail.com

OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

- Heart disease is one of the leading causes of death worldwide. Several factors such as age, lifestyle habits, cholesterol levels, blood pressure, and stress levels contribute to the risk of heart attacks.

- Healthcare organizations collect large amounts of patient health data, but analyzing this data manually to identify risk patterns can be difficult and time-consuming.

- There is a need for an interactive system that can analyze patient health data and visualize the relationships between different medical and lifestyle factors that contribute to heart attack risk.

PROPOSED SOLUTION:

The proposed system develops a Heart Attack Risk Analysis Dashboard using Power BI to analyze patient health data and identify key risk factors.

The system performs the following tasks:

- **Data Analysis:** Patient data including age, cholesterol, BMI, blood pressure, smoking habits, and stress levels are analyzed.
- **Interactive Dashboard:** The system visualizes patient health information through charts, graphs, and dashboards.
- **Risk Pattern Identification:** The dashboard highlights relationships between health factors and heart attack risk.
- **Decision Support:** The analysis helps researchers and healthcare analysts better understand patient health patterns.

SYSTEM APPROACH:

The project is developed using data visualization and analytics tools.

- **Tools Used:** Power BI Desktop
- **Dataset Source:** Heart Attack Prediction Dataset obtained from Kaggle.

The dataset includes patient health attributes such as:

- Age
- Gender
- Cholesterol
- Blood Pressure
- BMI
- Smoking
- Exercise Hours
- Sleep Hours
- Stress Level
- Heart Attack Risk Indicator

Data Processing:

The dataset is cleaned and prepared before analysis. This includes:

- Checking data types
- Creating calculated columns
- Creating DAX measures
- Removing inconsistencies

ALGORITHM & DEPLOYMENT:

This project focuses on data analytics and visualization rather than predictive modeling.

- **Data Input**
  - The dataset contains patient health information used to analyze risk patterns.
- **Analysis Process**
  - Data is imported into Power BI and analyzed using DAX calculations and visual analytics techniques.
  - Measures such as total patients and high-risk patients are calculated.
- **Visualization**
  - Several charts are used for analysis including:
  - Smoking vs Heart Attack Risk
  - Exercise vs Risk
  - Cholesterol vs Heart Rate
  - BMI vs Risk
  - Blood Pressure Distribution
- **Deployment**

  The dashboard can be shared using:
  - Power BI Desktop
    Power BI Service for online reporting

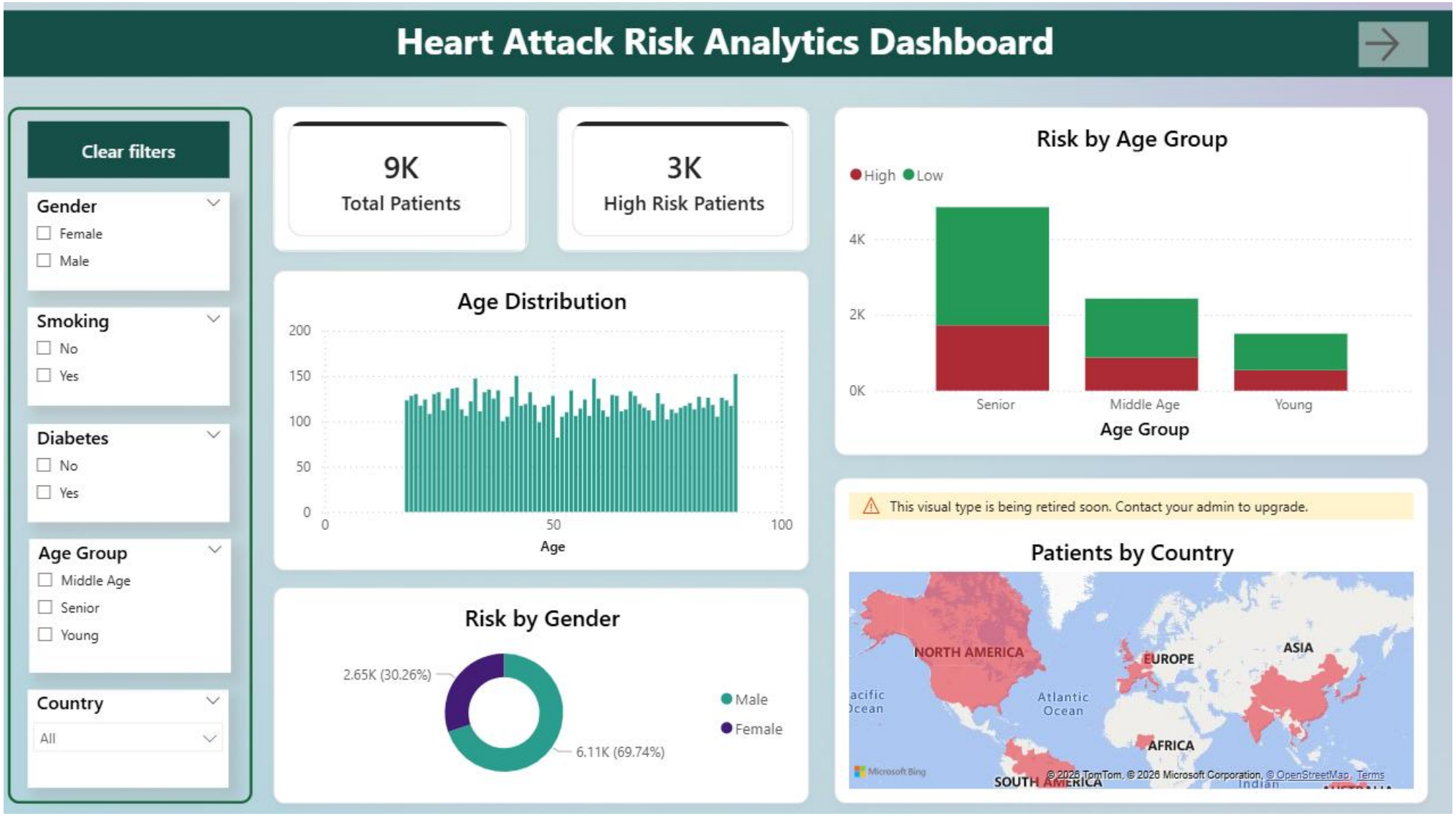
RESULT:

The dashboard successfully visualizes important health factors related to heart attack risk.

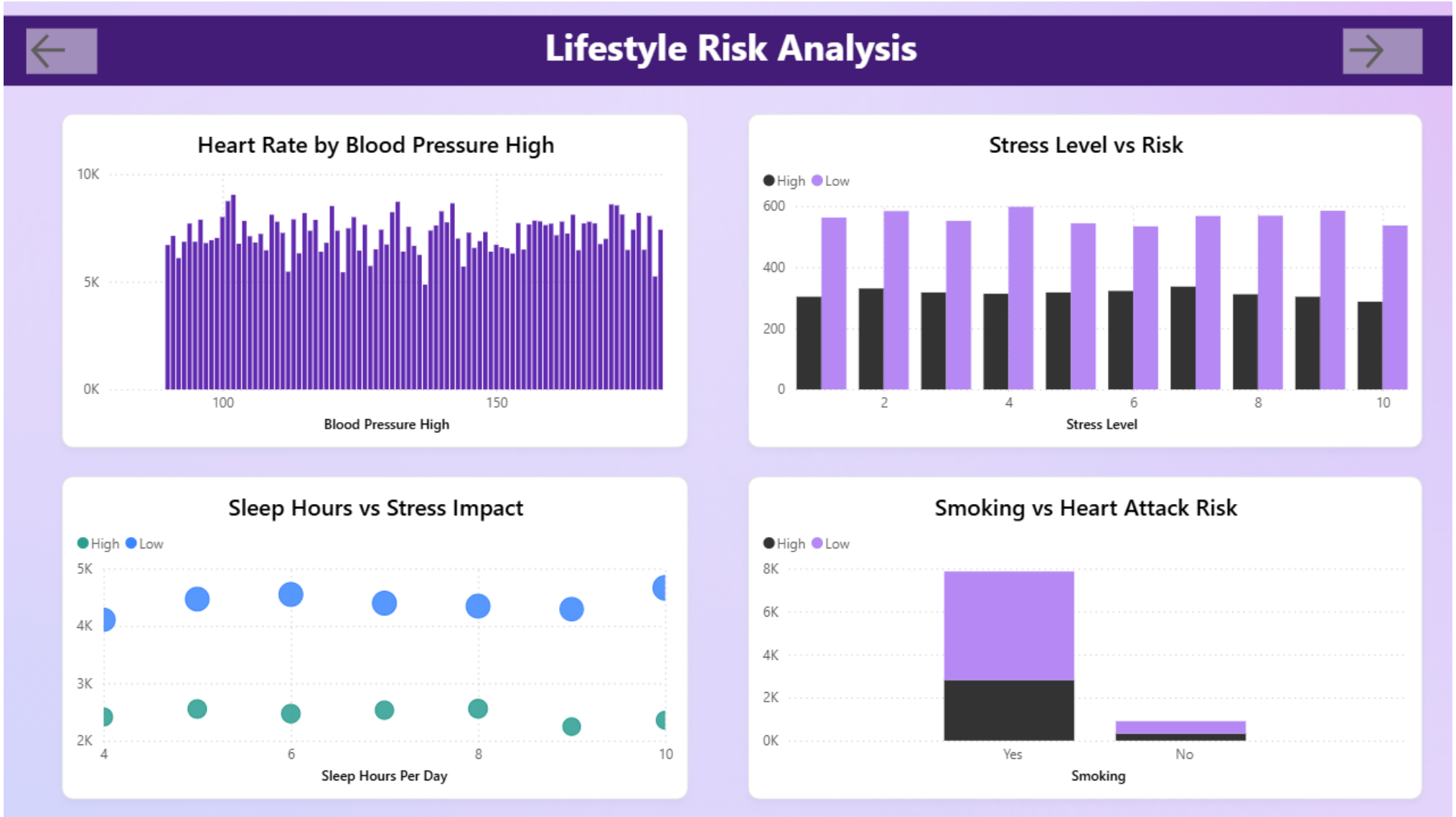
- Key insights obtained from the dashboard include:
  - High cholesterol levels are associated with increased heart attack risk.
  - Smoking habits significantly increase the likelihood of heart disease.
  - Patients with higher stress levels show greater risk patterns.
  - Overweight and obese BMI categories have higher heart attack probability.

The interactive dashboard allows users to explore patient health data using filters and visual analysis.

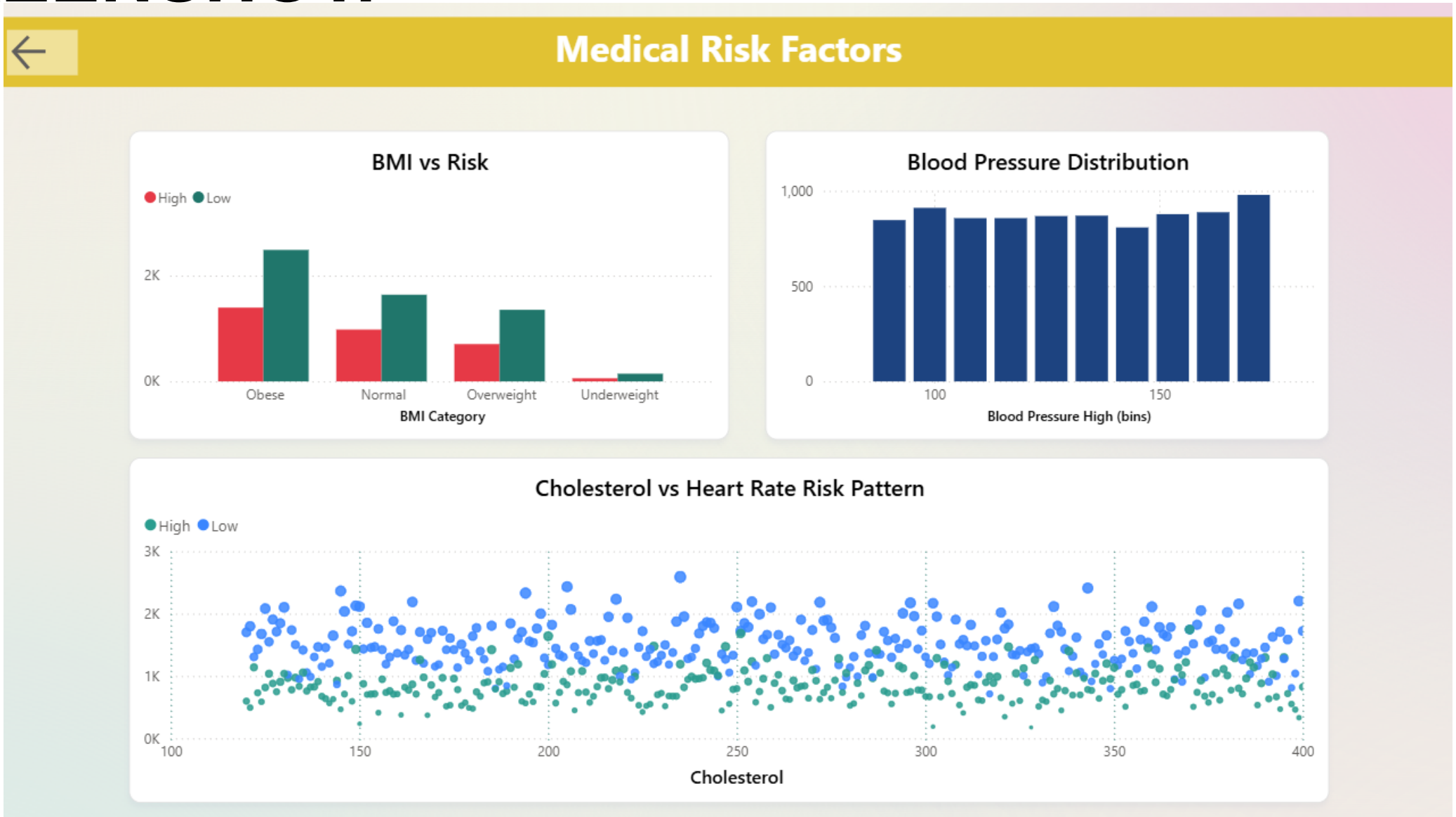
SCREENSHOT:


Heart Attack Risk Analytics Dashboard
Clear filters
Gender
Female
Male
Smoking
No
Yes
Diabetes
No
Yes
Age Group
Middle Age
Senior
Young
Country
All
9K
Total Patients
3K
High Risk Patients
Age Distribution
200
150
100
50
0
0
50
100
Age
Risk by Gender
2.65K (30.26%)
6.11K (69.74%)
Male
Female
Risk by Age Group
High
Low
4K
2K
0K
Senior
Middle Age
Young
Age Group
This visual type is being retired soon. Contact your admin to upgrade.
Patients by Country
NORTH AMERICA
EUROPE
ASIA
Atlantic Ocean
AFRICA
SOUTH AMERICA
Microsoft Bing
© 2026 TomTom, © 2026 Microsoft Corporation, © OpenStreetMap, Terms

SCREENSHOT:


Lifestyle Risk Analysis
Heart Rate by Blood Pressure High
10K
5K
0K
100
150
Blood Pressure High
Stress Level vs Risk
High
Low
600
400
200
0
2
4
6
8
10
Stress Level
Sleep Hours vs Stress Impact
High
Low
5K
4K
3K
2K
4
6
8
10
Sleep Hours Per Day
Smoking vs Heart Attack Risk
High
Low
8K
6K
4K
2K
0K
Yes
No
Smoking

SCREENSHOT:


Medical Risk Factors
BMI vs Risk
High
Low
2K
0K
Obese
Normal
Overweight
Underweight
BMI Category
Blood Pressure Distribution
1,000
500
0
100
150
Blood Pressure High (bins)
Cholesterol vs Heart Rate Risk Pattern
High
Low
3K
2K
1K
0K
100
150
200
250
300
350
400
Cholesterol

CONCLUSION:

- The Heart Attack Risk Analysis Dashboard provides an effective way to analyze healthcare data and identify important health risk factors.
- By transforming complex datasets into clear visualizations, the dashboard helps users understand how different medical and lifestyle factors influence heart attack risk.
- This project demonstrates how data analytics tools like Power BI can support healthcare data analysis and improve understanding of disease risk patterns.

FUTURE SCOPE:

The project can be improved in several ways:

- Integrating larger healthcare datasets
- Adding real-time health monitoring data
- Developing predictive models for heart attack risk
- Creating a web-based healthcare analytics system
- Expanding analysis for multiple diseases

These improvements could enhance the system's ability to support healthcare research and decision-making.

REFERENCES:

- Kaggle Dataset
  Heart Attack Prediction Dataset
  https://www.kaggle.com/datasets/iamsouravbanerjee/heart-attack-prediction-dataset
- Microsoft Power BI Documentation
  https://learn.microsoft.com/power-bi/
- Healthcare analytics research articles.

GitHub Link:Github Link

Thank You